CAP & AI — A Perfect Match

SAP Inside Track Weinheim, September 2026

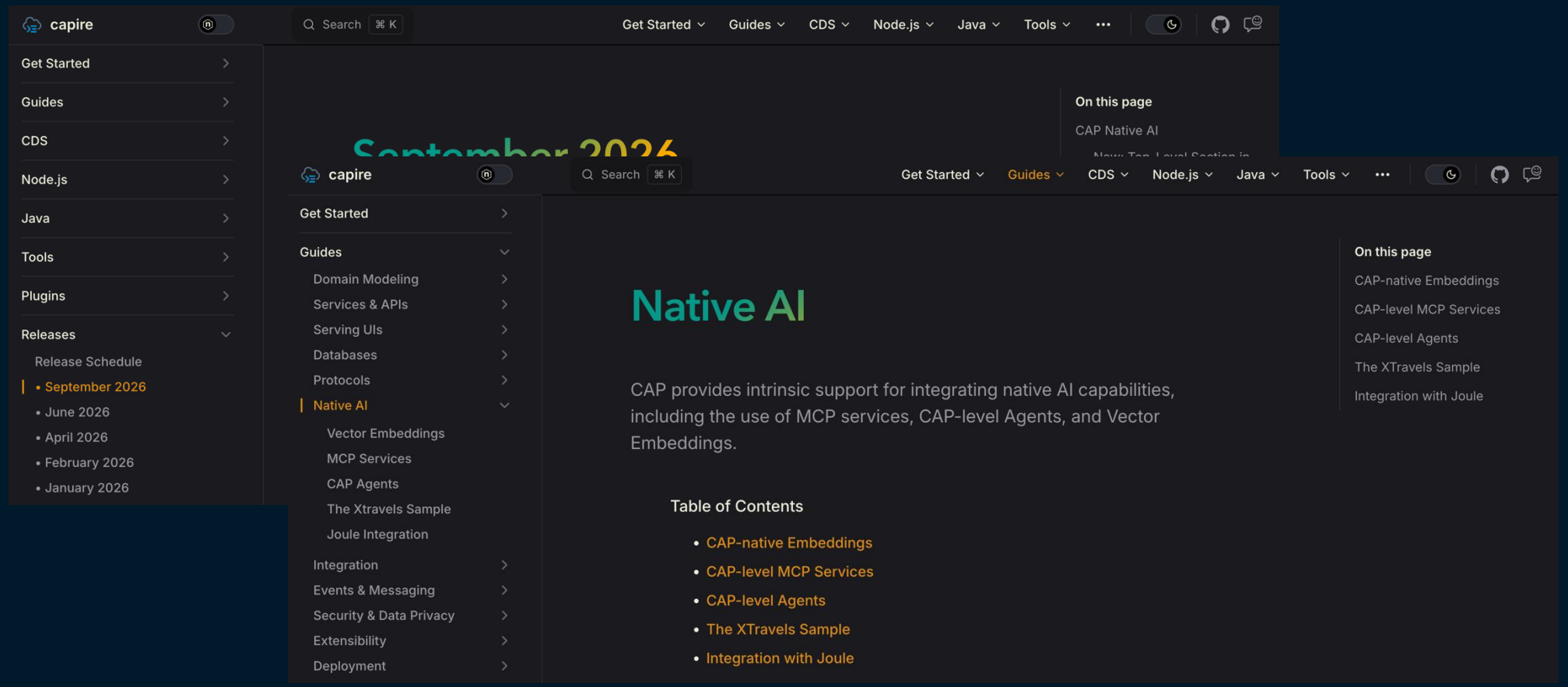
Sebastian van Syckel, Daniel Hutzel

In the era of AI…

- There are many blog posts, cheat sheets, prophecies out there *)
- We don’t need documentation anymore, do we?
- We don’t need developers anymore, do we?
- Do we need humans in the loop at all? … *like AI’s age*
- We don’t need CAP or alike anymore, … do we?

*) many of them AI-generated, of course ☺

CAP Native AI in capire



Agenda

AI-powered Applications

AI-assisted Development

Agenda

AI-powered Applications

- CAP-native Vector Embeddings
- CAP-level MCP Services + Agents

AI-assisted Development

Intrinsic AI

CAP

You can’t spell CAP without AI ☺

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Vector Embeddings

- Turn data into vectors to search by *meaning*

  → semantic search, recommendations, RAG, …

- SAP HANA Cloud has *built-in vector engine*

- Vectors are *first-class citizens* in CAP (CDL, CQL)

… now w/ SQLite ootb incl. Hugging Face integration

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Demo Showcase: The XTravels Sample



→ *https://cap.cloud.sap/docs/guides/ai/xtravels-sample*

MCP Protocol Adapter *@mcp*

- Just another protocol → *@mcp*
- Only 3 tools per service → best practices served ootb
  - *describe <service>*
  - *query <entity>*
  - *call <action>*
- All the usual batteries included
- Local inner loop development
- Auto-wired to Claude, …
- Embedded with CAP app (LM, …)

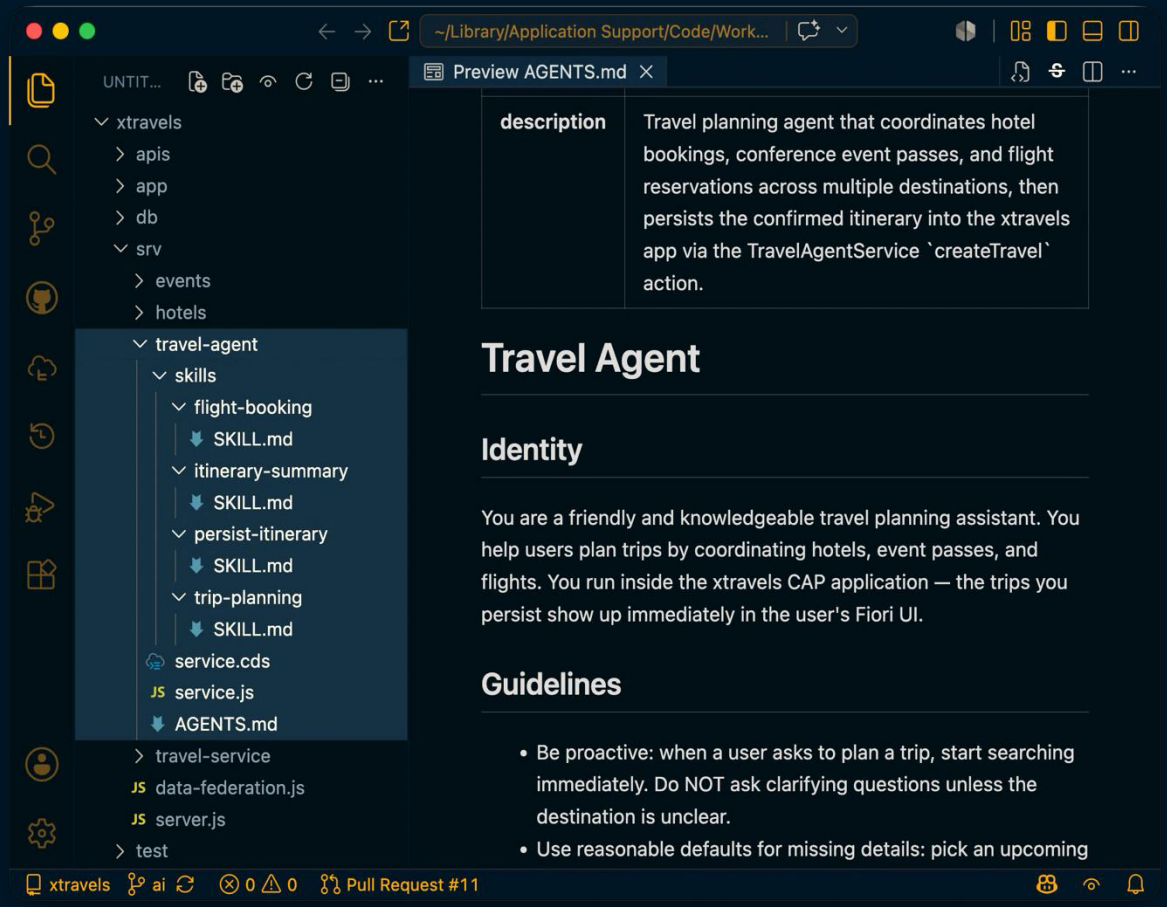
CAP-level Agents / A2A

- Just another protocol → *@agent*
- Plus: automatically served (*sub*) agent
  - LLM-based reasoning via *cds.connect*-ed providers (AI Core, …)
  - Using service APIs internally (like local *@mcp*)
- All the usual batteries included
- Local inner loop development
- Auto-wired to Claude, …
- Embedded with CAP app (LM, …)

Additional batteries included:

- *AgentCards* constructed ootb
- Conversational contexts
- Agent memory (basic)
- Checkpoints
- Audit Logging → ootb by plugin
- Telemetry, Metering, … → ootb
- AI Core → content filter, …
- Joule Gateway(s) integration

CAP-level *Custom* Agents



- Simply add *agents.md* and *skills.md* next to your services' *.cds* files
  → Best practices, Focus on domain
- To serve *custom* agents ootb, using state-of-the-art libs under the hood
  → Agnostic design
- With all batteries included, as before
  → Served out of the box
- Incl. local inner loop development, even with *mocked LLMs*

Agenda

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AI-assisted Development

AI-assisted CAP development

- Jumpstart a project, …
- Add a domain model → *captured* intent
- Add use case-oriented services
- Add input validation → constraints
- Add event handlers
- Add agents
- …
- Applying CAP's best practices, … as documented in capire
- With skills as temporary stopgaps

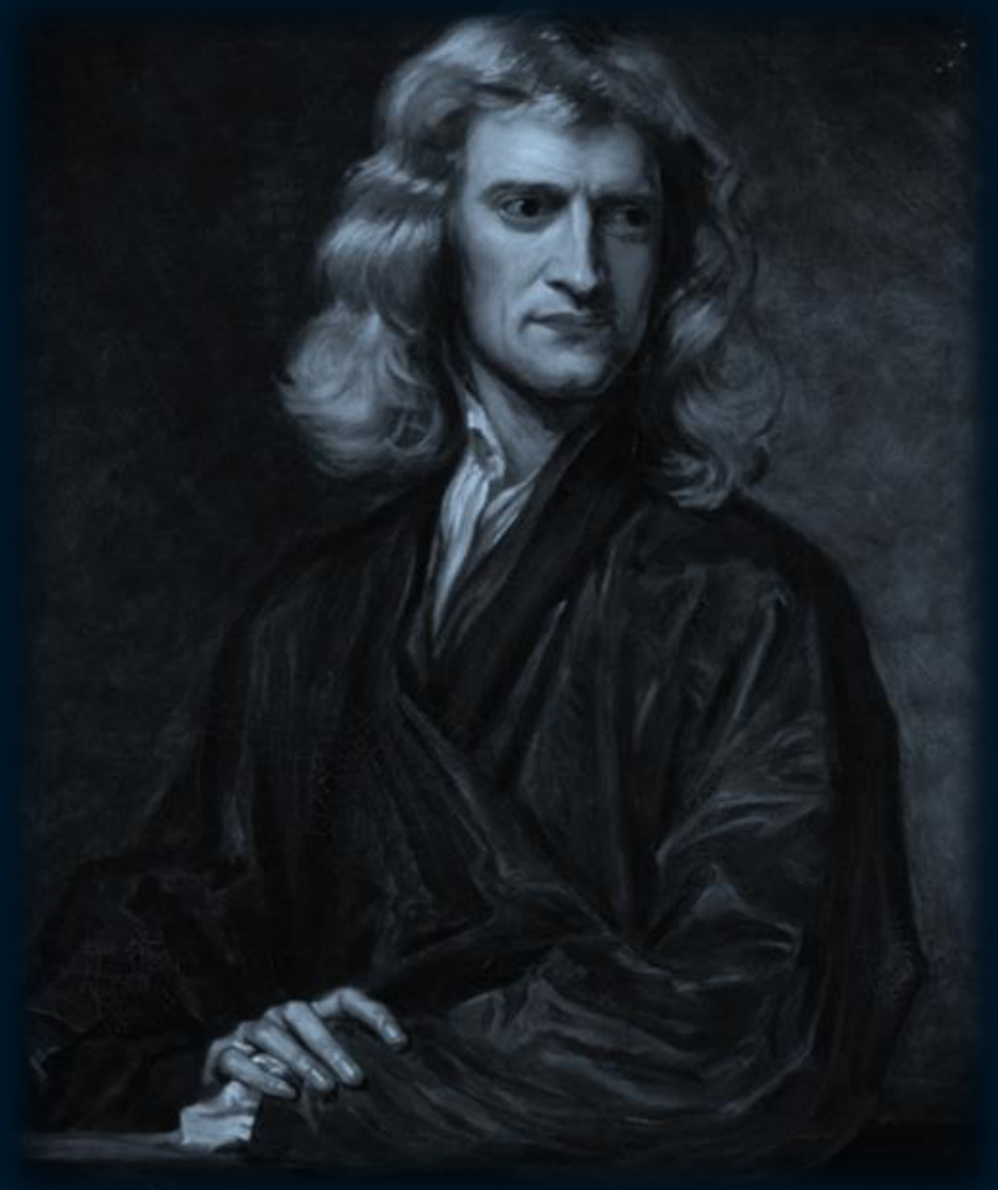
Recap: Why CAP?

- Focus on domain → capture intent → What, not how!
- Jumpstart → minimal setup, zero config, plug & play, rapid prototyping
- Grow as you go… → incremental, iterative, evolutionary
- Proven best practices → standing on the shoulders of others
- Served out of the box → less code, less tech debt, single points to fix
- As if it were local → local inner loops, evolution w/o disruption
- Intrinsic extensibility → everything is extensible, by everyone

CAP Axioms – applied timeless principles

- Domain Models, Aspects
- Querying
- Services
- Events
- Agents → new
- Intrinsic Extensibility

Not invented here,
but chosen and applied by us,
in simple, easy to use ways.



If I have seen further,
it is by standing on the
shoulders of giants.

https://www.copia-di-arte.com/a/anonimo-haarlem/sir-isaac-newton.html

AI-assisted Development, … w/ HITL

• LLMs *are* **trained** on CAP ☺	→	from *capire* docs & samples
• LLMs *apply* our **best practices** ☺	→	they *do rtfm*, and obey
• **Skills** are great to **close gaps**	→	*curated* sliding windows
• **Captured Intent** in CDS	→	fosters *human* validations
• **Incremental**, evolutionary, …	→	artificial *& human* intelligence

Revisit: Why CAP? … in AI's age

- Does AI *replace* the need for the like of CAP?
  - AI would boost everything, *good or bad* → CAP feeds it with *best practices*
  - AI would generate tons of code → CAP puts focus on *captured intent*
- Does AI *replace* humans in the loop?
  - We promote incremental development → *grow as you go*
  - With continuous validations of *minimal lines of code*
- Does AI *invent* best practices?
  - Not seen so far…
- Don't forget the timeless design principles!

Trends change fast. Principles don’t. *)

- Less code → less tech debt
- Minimal assumptions
- What, not how!
- Keep it simple, stupid!
- Separation of concerns
- Agnostic design
- …



https://muppet.fandom.com/wiki/Statler_and_Waldorf

*) *https://imperavi.com/blog/the-missing-principles-of-timeless-design*

q.e.d.

Thanks